

Inspectable Symbolic Compression for Long-Context Reasoning: A Mixed-Results Study

Anonymous authors - English short-paper preview generated from the local study report

Abstract

We study Inspectable Symbolic Compression (ISM), a prompt-only method that rewrites long documents as symbols, dictionary definitions, and relations. The result is mixed. ISM structure is functionally used: semantic dictionary damage and symbolic-structure removal both reduce accuracy. Yet a natural-language model summary is significantly more accurate and more token-efficient under the same budgets, and it dominates reuse cost-accuracy. We therefore do not claim prompt-only ISM is a better compressor. The study identifies a boundary: symbolic text written by an LLM and read by an LLM remains in the same class as summarization. Future work should treat ISM as executable semantic IR parsed and checked outside the LLM loop.

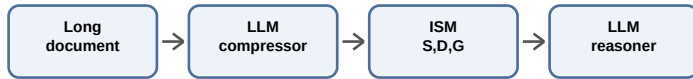


Figure 1. Prompt-only ISM: the LLM writes a structured text, and an LLM reads it again.

1. Introduction

Prompt-compression work usually optimizes token count and retained downstream accuracy. We ask a narrower diagnostic question: can a compressed representation expose meaning units that can be inspected and intervened on? ISM forces the compressor to emit symbols, a short dictionary, and relations such as precedence or exception.

The original hypothesis was that this symbolic text would be both compact and manipulable. The experiments preserve only the second part. ISM is read and used by the model, but it does not beat ordinary summaries as a prompt-only compressor.

2. Related Work

Token-level prompt compression. LLMingua, LongLLMLingua, LLMLingua-2, and Selective Context reduce context by selecting, reweighting, or dropping tokens. These methods are strong baselines for budgeted prompting, but the compressed object is still an optimized prompt span.

Learned memory compression. Gist Tokens, AutoCompressors, and ICAE replace long context with trained latent or soft-token memory. They can be very compact, but their internal state is not directly inspectable or locally editable by a human or a symbolic checker.

Intervenable bottlenecks. Concept Bottleneck and Concept Embedding Models show why an intermediate representation can be valuable even when it is not the shortest one: the user can inspect or intervene on named concepts. ISM imports that idea into prompt compression, but keeps the representation textual and prompt-only.

3. Method

Each example contains a document x_i , questions q_{ij} , and labels y_{ij} . A producer m writes a compressed context z_i^m under budget B , and the same reasoner answers from that context.

$$z_i^m(B) = P_m(x_i; B), \text{tokens}(z_i^m) \leq B \\ \hat{y}_{ij}^m = R(z_i^m(B), q_{ij})$$

ISM is the producer whose output is $z=(S,D,G)$: symbol labels $S=\{Z_1,\dots,Z_k\}$, dictionary entries D mapping each label to a condition and conclusion, and relations G encoding precedence, exception, or conjunction structure. Natural-language summary, keyword extraction, and oracle gold summary are evaluated under the same answer protocol.

$$\text{Acc}_m = \text{mean}[1(\hat{y}_{ij}^m = y_{ij})] \\ \text{AR}_m = \text{Acc}_m / \text{Acc}_{\text{full}}; \text{CR}_m = \text{tokens}(z_m) / \text{tokens}(x) \\ \text{ES}_m = \text{AR}_m / \text{CR}_m; \Delta_{\text{flip}} = \text{Acc}_{\text{full_dict}} - \text{Acc}_{\text{flipped}}$$

Interventions separate three constructs. Derangement permutes labels while preserving the definition multiset; conclusion flip changes the semantic content of dictionary rules; blanking removes dictionary content; symbol-only and random-symbol controls test whether relations carry information beyond arbitrary markers.

4. Results

RQ1: ISM structure is used. After correcting a low-purity compressor and replacing weak label derangement with semantic conclusion flips, dictionary content and symbolic structure both show functional effects.

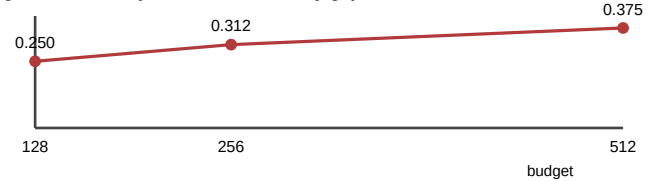
Condition	Acc	AR	CR
Full Context	0.750	1.000	1.000
Full Symbol+Dict	0.446	0.594	0.745
Flipped Dict	0.367	0.489	0.745
Symbol Only	0.413	0.550	0.079
Random Symbol	0.304	0.406	0.738

Dev scale-up: $\Delta_{\text{map_flip}}=+0.079$ (95% CI [0.013,0.146], $p=0.032$); $\Delta_{\text{symbol}}=+0.108$ (95% CI [0.050,0.167], $p=5e-4$). Derangement is non-significant.

RQ3: summaries beat ISM under fixed budgets. The primary analysis is paired on the same 80 questions, so it is not explained by full-context filler degradation.

Budget	Summary-ISM	95% CI	p
128	+0.2500	[0.100,0.400]	0.0029
256	+0.3125	[0.175,0.450]	1.1e-4
512	+0.3750	[0.250,0.500]	2.3e-7

Figure 2. Summary minus ISM accuracy gap



Fixed-budget AR/ES cells show the same pattern. At budgets 128/256/512, Model Summary reaches 1.036/51.3, 1.125/17.3, and 1.196/13.3, while ISM reaches 0.679/9.4, 0.679/9.3, and 0.661/8.9.

RQ4: reuse is not ISM-specific. Cached representations become cheaper than full context from $n=2$ questions onward, but Model Summary is cheaper and more accurate than ISM.

Method	n=1	n=8	n=64	Acc
Full Context	1354	10828	86624	0.700
Model Summary	1526	2168	7305	0.787
ISM	1548	2265	8005	0.475
Keyword	1820	3492	16868	0.512

Takeaway. The prompt-only format gives a real diagnostic signal but loses the efficiency contest. A stronger future paper should move from LLM-readable text to program-readable semantic IR.

5. Discussion

The positive result is that ISM is not inert: the model uses semantic dictionary content and symbolic relations. The negative result is stronger for the original compression claim: in the prompt-only setting, ISM and Model Summary are both LLM-written texts read by an LLM, and the ordinary summary is the better text cache.

Thus the study should not be read as a successful new compressor. It is a boundary result. To become categorically different from summaries, ISM must be parsed, checked, transformed, and executed by programs outside the LLM. The future claim is not 'summaries cannot be edited' but 'semantic IRs permit automatic, local, schema-validated edits and symbolic execution.'

6. Limitations

The reported experiments are dev-scale and synthetic. QASPER and LLMLingua-2 are specified but not fully evaluated here. Several large GPU artifacts are stored by sha256 and reproducibility commands. RQ2 dictionary swap remains future work. Most importantly, this paper evaluates prompt-only ISM, not a parser/checker/executor system.

7. Conclusion

ISM representations are functionally used, but prompt-only ISM is not a better compressor than natural-language summaries. This mixed result closes one path and clarifies the next: executable semantic IR for long-context reasoning.

The boundary claim is therefore precise: structured prompt text can expose units that an LLM uses, but it does not by itself create a new computational object. The next version must make the representation parseable, checkable, and executable by ordinary software.

8. Pre-registered Status

Criterion	Status	Evidence
#1 ISM > Summary	Failed	RQ3/RQ4 summary dominates
#2 structure used	Met	flip p=0.032; symbol p=5e-4
#3 unseen swap	Open	LoRA swap not run

By the original decision rule, failure of criterion #1 means the paper cannot claim an efficiency advantage for ISM. Criterion #2 supports functional use of the internal structure, but that is distinct from being a better compressor.

9. Semantic-IR Path

Property	Summary	Semantic IR
Format	free text	schema / AST
Validation	manual	checker
Edit	rewrite	rule-id local edit
Execution	LLM	symbolic executor

The next research question is compile-and-execute: can an LLM compile a long document into an IR that a non-LLM program can parse and run? That would make the representation categorically different from a short natural-language summary.

10. Evidence Ledger

Finding	Evidence	Status
ISM is used	ablation-qwen7b-N120	positive
Summary wins	fixed-budget-N40	negative
Reuse not unique	reuse-N40	negative
Root cause	llm-ism-diagnostic	resolved

The evidence pattern is internally consistent: the representation carries signal, but the prompt-only implementation loses the efficiency contest to a plain summary.

11. Next Tests

Compile-and-execute. Convert ISM text into a schema-checked AST and answer with a deterministic executor. **Targeted edits.** Repair or delete a rule by id and verify the prediction changes locally. **Swap.** Finish the LoRA dictionary-swap experiment for unseen labels.

12. Reproducibility

Evidence is stored under docs/evidence/: ablation-qwen7b-N120 for RQ1, fixed-budget-N40 for RQ3, reuse-N40 for RQ4, and llm-ism-diagnostic.md for the corruption/purity diagnosis. Large GPU outputs are retained by sha256 and reproducibility commands.

References

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